



Laboratorium
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Laporan Akhir

Praktikum Jaringan Komputer

Vlan–Trunk–OSPF–MultiVendor

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1 Langkah-Langkah Percobaan

Bagian ini menjelaskan secara rinci tahapan dan prosedur yang dilakukan selama praktikum pada modul 5, meliputi konfigurasi Cisco Switch, Cisco Router, Huawei Router, serta konfigurasi OSPF multivendor dan pengujian failover OSPF.

1.1 Praktikum 1 — Konfigurasi Cisco Switch

1. Masuk ke terminal Cisco Switch, kemudian aktifkan mode privileged dan masuk ke mode konfigurasi global.

```
1 Switch> enable
2 Switch# configure terminal
3
```

2. Buat VLAN 10, 20, 30, dan 40 beserta nama masing-masing.

```
1 vlan 10
2   name Finance
3 exit
4 vlan 20
5   name HR
6 exit
7 vlan 30
8   name IT
9 exit
10 vlan 40
11   name Marketingg
12 exit
13
```

Verifikasi dengan perintah berikut untuk memastikan semua VLAN sudah aktif.

```
1 show vlan brief
2
```

```
Switch#show vlan brief

VLAN Name                Status    Ports
----
1    default              active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
10   FINANCE              active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                   active
30   IT                   active
40   MARKETING            active
1002 fddi-default         act/unsup
1003 token-ring-default  act/unsup
1004 fddinet-default     act/unsup
1005 trnet-default       act/unsup
Switch#
Switch#
```

Gambar 1: Hasil show vlan brief pada Cisco Switch

3. Konfigurasi *access port* untuk setiap VLAN sesuai interface yang terhubung ke masing-masing PC.

```
1 interface gi0/2
2   description PC-FINANCE-VLAN10
```

```

3  switchport mode access
4  switchport access vlan 10
5  no shutdown
6  exit
7
8  interface gi0/1
9      description PC-HR-VLAN20
10     switchport mode access
11     switchport access vlan 20
12     no shutdown
13 exit
14
15 interface gi1/1
16     description PC-IT-VLAN30
17     switchport mode access
18     switchport access vlan 30
19     no shutdown
20 exit
21
22 interface gi0/3
23     description PC-MARKETING-VLAN40
24     switchport mode access
25     switchport access vlan 40
26     no shutdown
27 exit
28

```

4. Konfigurasi *trunk port* pada interface Gi0/0 yang terhubung ke Cisco Router untuk membawa VLAN 10, 20, 30, dan 40.

```

1  interface gi0/0
2      switchport trunk encapsulation dot1q
3      description TRUNK-TO-CISCO-ROUTER
4      switchport mode trunk
5      switchport trunk allowed vlan 10,20,30,40
6      no shutdown
7  exit
8

```

Verifikasi trunk dengan perintah berikut.

```

1  show interfaces trunk
2

```

```
*Jun 6 03:33:49.589: %SYS-5-CONFIG_I: Configured from console
Switch#
Switch#show interfaces trunk

Port      Mode      Encapsulation  Status      Native vlan
Gi0/0     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gi0/0     10,20,30,40

Port      Vlans allowed and active in management domain
Gi0/0     10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gi0/0     10,20,30,40
Switch#
Switch#
```

Gambar 2: Hasil show interfaces trunk pada Cisco Switch

5. Simpan semua konfigurasi switch dengan perintah berikut.

```
1 write memory
2
```

1.2 Praktikum 2 — Konfigurasi Cisco Router

1. Masuk ke Cisco Router dan aktifkan mode konfigurasi global.

```
1 enable
2 configure terminal
3
```

2. Konfigurasi interface fisik Gig0/2 yang terhubung ke Cisco Switch sebagai *trunk parent interface* tanpa IP address.

```
1 interface gig0/2
2 description TRUNK-TO-CISCO-SWITCH
3 no ip address
4 no shutdown
5 exit
6
```

3. Konfigurasi subinterface untuk setiap VLAN (ROAS — *Router on a Stick*) beserta DHCP server untuk VLAN 10 dan 20.

```
1 interface gig0/2.10
2 description GATEWAY-VLAN10-FINANCE
3 encapsulation dot1Q 10
4 ip address 192.168.10.1 255.255.255.0
5 no shutdown
6 exit
7
8 interface gig0/2.20
9 description GATEWAY-VLAN20-HR
10 encapsulation dot1Q 20
11 ip address 192.168.20.1 255.255.255.0
12 no shutdown
```

```

13 exit
14
15 interface gig0/2.30
16   description GATEWAY-VLAN30-IT
17   encapsulation dot1Q 30
18   ip address 192.168.30.1 255.255.255.0
19   no shutdown
20 exit
21
22 interface gig0/2.40
23   description GATEWAY-VLAN40-MARKETING
24   encapsulation dot1Q 40
25   ip address 192.168.40.1 255.255.255.0
26   no shutdown
27 exit
28

```

4. Konfigurasi DHCP server untuk VLAN 10 dan VLAN 20.

```

1 ip dhcp excluded-address 192.168.10.1 192.168.10.10
2 ip dhcp pool VLAN10-FINANCE
3   network 192.168.10.0 255.255.255.0
4   default-router 192.168.10.1
5   dns-server 8.8.8.8
6 exit
7
8 ip dhcp excluded-address 192.168.20.1 192.168.20.10
9 ip dhcp pool VLAN20-HR
10  network 192.168.20.0 255.255.255.0
11  default-router 192.168.20.1
12  dns-server 8.8.8.8
13 exit
14

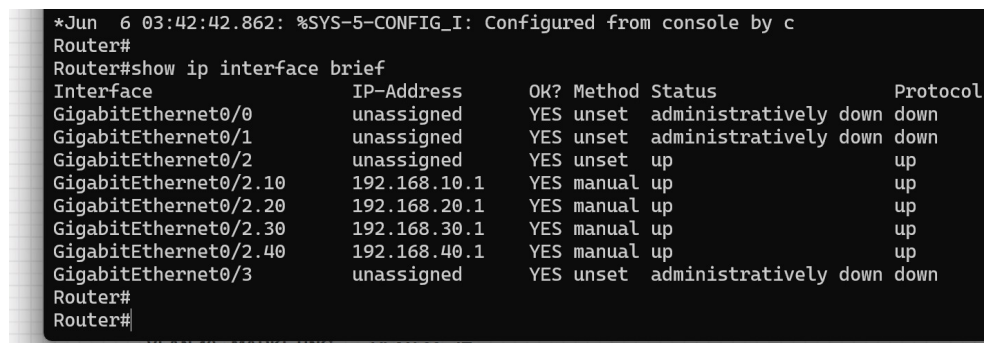
```

Verifikasi interface Cisco Router untuk memastikan subinterface sudah aktif.

```

1 show ip interface brief
2

```



```

*Jun  6 03:42:42.862: %SYS-5-CONFIG_I: Configured from console by c
Router#
Router#show ip interface brief
Interface                                IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0                       unassigned      YES unset  administratively down  down
GigabitEthernet0/1                       unassigned      YES unset  administratively down  down
GigabitEthernet0/2                       unassigned      YES unset  up                    up
GigabitEthernet0/2.10                    192.168.10.1    YES manual  up                    up
GigabitEthernet0/2.20                    192.168.20.1    YES manual  up                    up
GigabitEthernet0/2.30                    192.168.30.1    YES manual  up                    up
GigabitEthernet0/2.40                    192.168.40.1    YES manual  up                    up
GigabitEthernet0/3                       unassigned      YES unset  administratively down  down
Router#
Router#

```

Gambar 3: Hasil show ip interface brief pada Cisco Router

Verifikasi DHCP pool yang telah dikonfigurasi.

```

1 show ip dhcp pool

```

```

Router#
Router#show ip dhcp pool

Pool VLAN10-FINANCE :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.10.12 192.168.10.1 - 192.168.10.254 1

Pool VLAN20-HR :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 254
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
192.168.20.12 192.168.20.1 - 192.168.20.254 1
Router#
Router#

```

Gambar 4: Hasil show ip dhcp pool pada Cisco Router

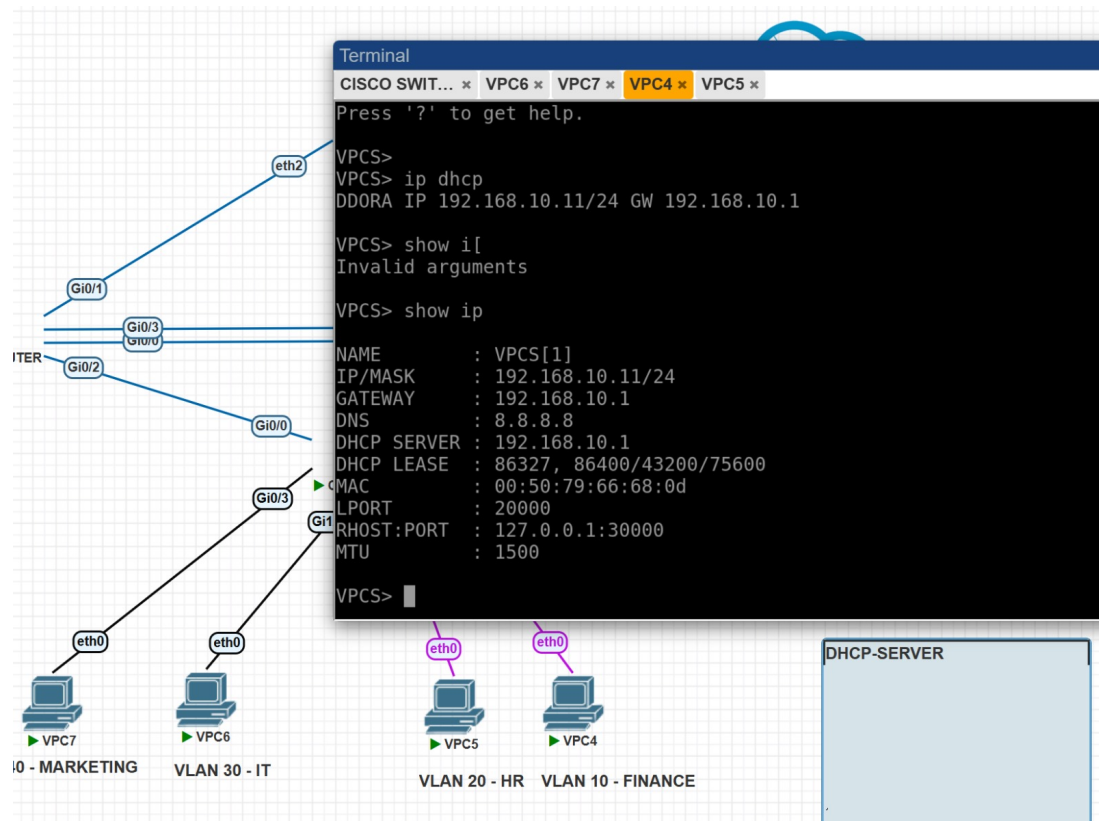
5. Konfigurasi IP address pada setiap VPCS. VLAN 10 dan 20 menggunakan DHCP, sedangkan VLAN 30 dan 40 menggunakan IP statis.

```

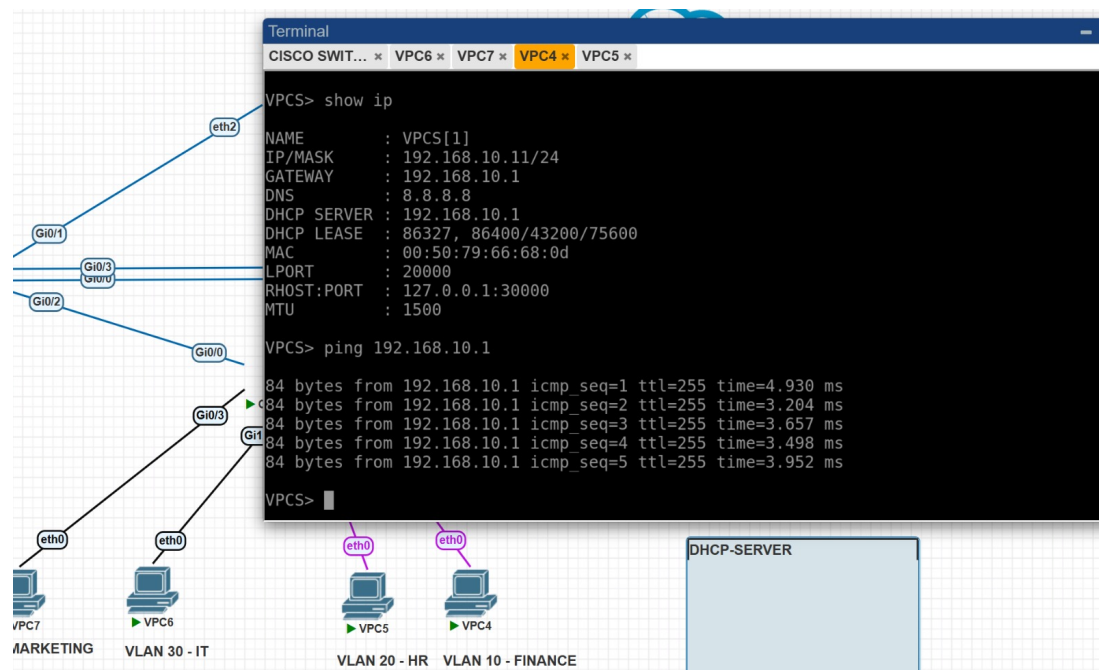
1 ! PC VLAN 10 dan 20
2 ip dhcp
3
4 ! PC VLAN 30
5 ip 192.168.30.10/24 192.168.30.1
6
7 ! PC VLAN 40
8 ip 192.168.40.10/24 192.168.40.1
9

```

6. Verifikasi bahwa PC VLAN 10 mendapatkan IP dari DHCP dan dapat melakukan ping ke gateway.

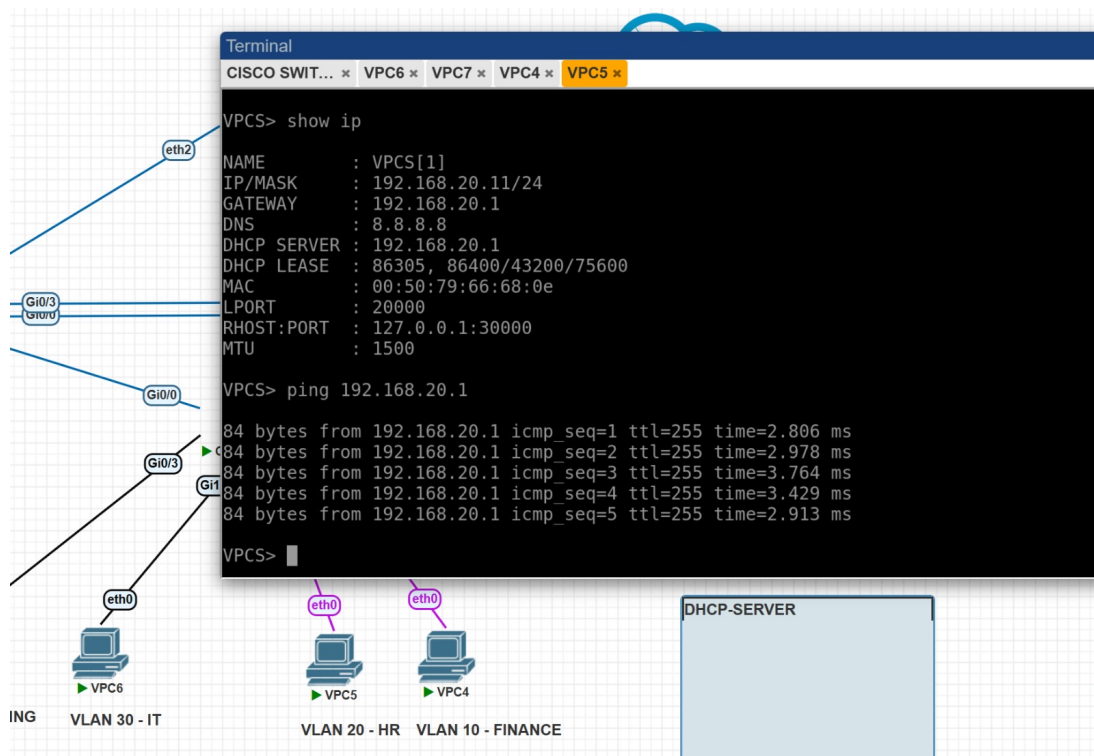


Gambar 5: PC VLAN 10 mendapat IP DHCP 192.168.10.11/24



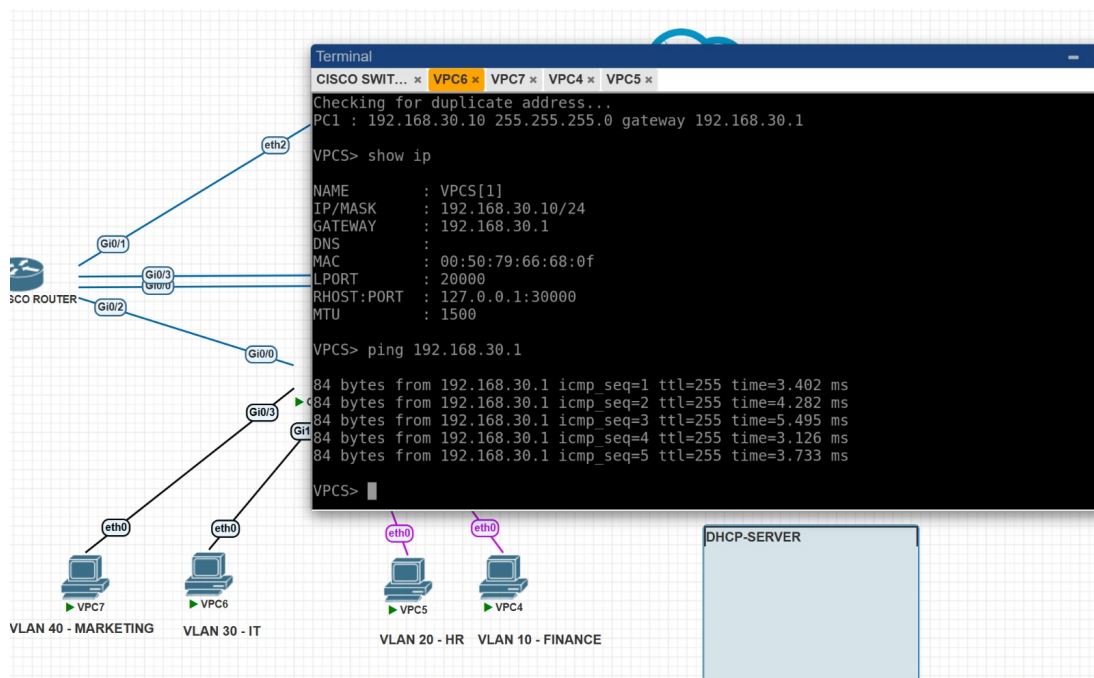
Gambar 6: Ping gateway 192.168.10.1 dari PC VLAN 10 berhasil

7. Verifikasi bahwa PC VLAN 20 mendapatkan IP dari DHCP dan dapat melakukan ping ke gateway.

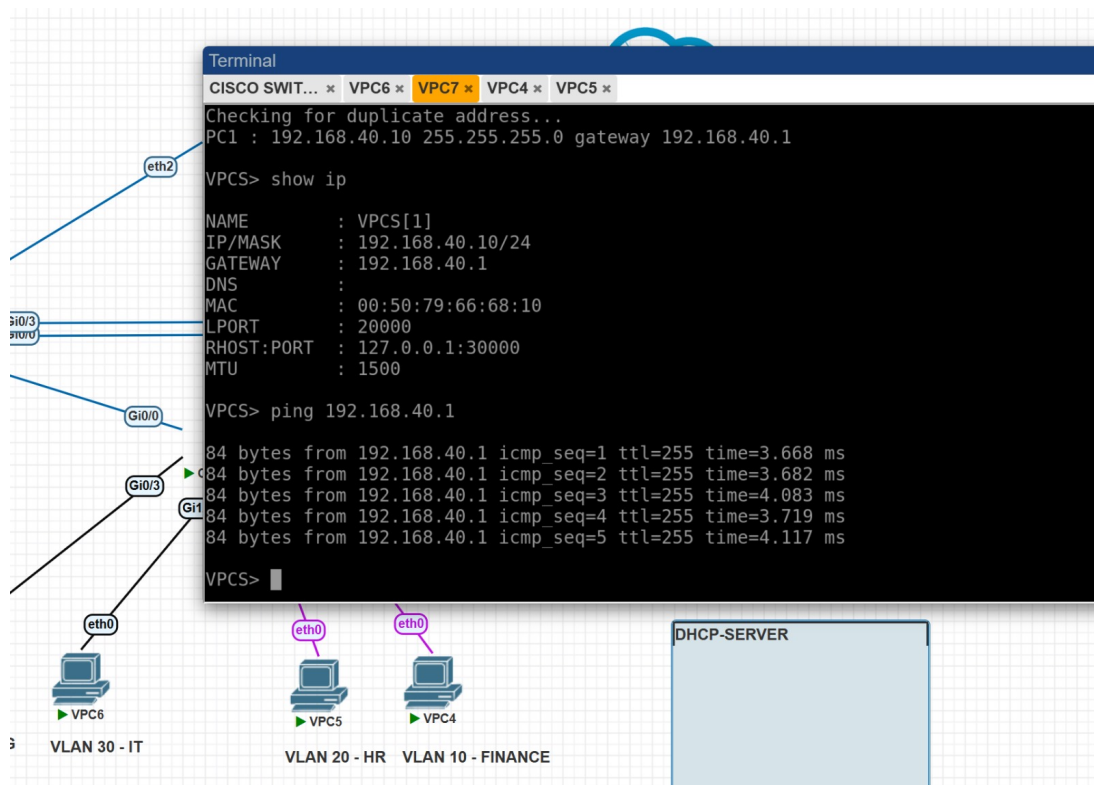


Gambar 7: PC VLAN 20 mendapat IP DHCP 192.168.20.11/24 dan ping gateway berhasil

8. Verifikasi PC VLAN 30 dan VLAN 40 dengan IP statis dapat melakukan ping ke gateway masing-masing.



Gambar 8: PC VLAN 30 (IP statis 192.168.30.10/24) ping gateway berhasil

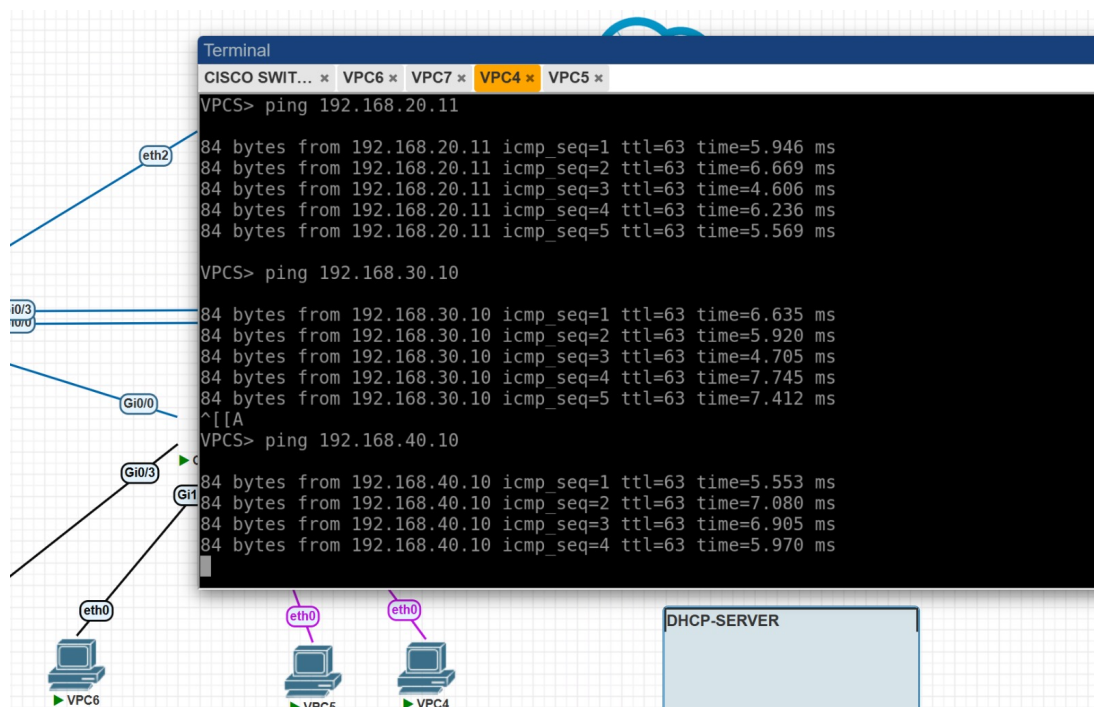


Gambar 9: PC VLAN 40 (IP statis 192.168.40.10/24) ping gateway berhasil

9. Uji konektivitas antar-VLAN dari PC VLAN 10 ke VLAN 20, 30, dan 40.

```

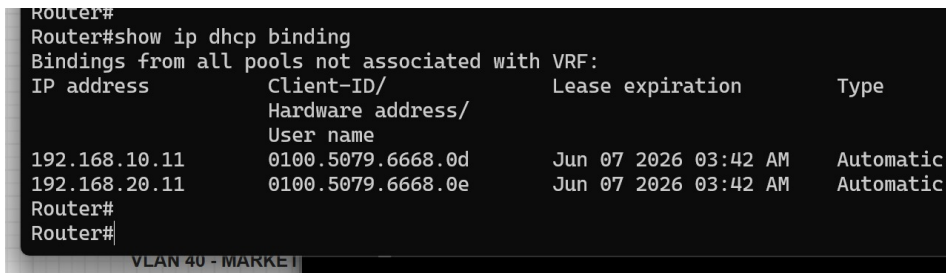
1 ping 192.168.20.11
2 ping 192.168.30.10
3 ping 192.168.40.10
4
  
```



Gambar 10: Ping antar-VLAN dari PC VLAN 10 ke VLAN 20, 30, dan 40

10. Verifikasi DHCP binding pada Cisco Router untuk memastikan IP yang diberikan sesuai dengan PC klien.

```
1 show ip dhcp binding
2
```



The screenshot shows the output of the 'show ip dhcp binding' command on a Cisco Router. The output lists two DHCP bindings with their IP addresses, client IDs, hardware addresses, lease expiration times, and types.

IP address	Client-ID/ Hardware address/ User name	Lease expiration	Type
192.168.10.11	0100.5079.6668.0d	Jun 07 2026 03:42 AM	Automatic
192.168.20.11	0100.5079.6668.0e	Jun 07 2026 03:42 AM	Automatic

Gambar 11: Hasil show ip dhcp binding pada Cisco Router

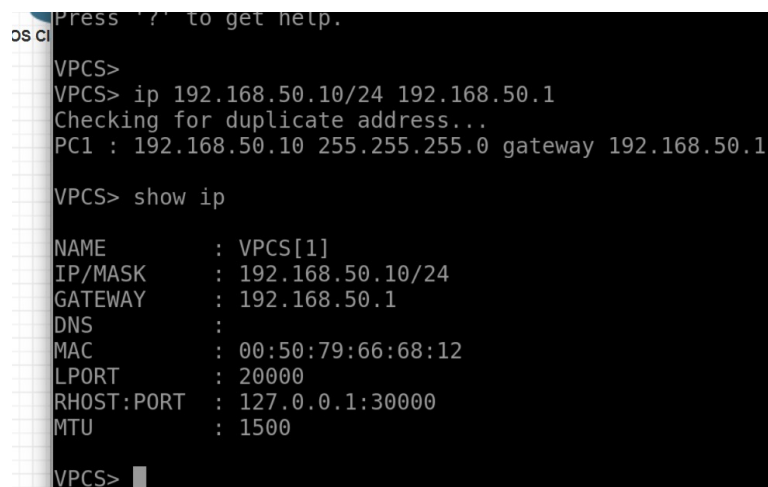
1.3 Praktikum 3 — Konfigurasi Huawei Router dan IP Address Antarrouter

1. Konfigurasi IP address untuk LAN Huawei pada interface Ethernet1/0/2 dengan network 192.168.50.0/24.

```
1 system-view
2
3 interface Ethernet1/0/2
4 description LAN-HUAWEI
5 ip address 192.168.50.1 255.255.255.0
6 undo shutdown
7 quit
8
9 commit
10
```

2. Konfigurasi IP statis pada PC LAN Huawei dan verifikasi konektivitasnya.

```
1 ip 192.168.50.10/24 192.168.50.1
2
```



The screenshot shows the terminal output of a Huawei VPCS (Virtual PC System) configuration. It displays the IP address configuration for a PC and the resulting network parameters.

```
Press '?' to get help.
VPCS>
VPCS> ip 192.168.50.10/24 192.168.50.1
Checking for duplicate address...
PC1 : 192.168.50.10 255.255.255.0 gateway 192.168.50.1

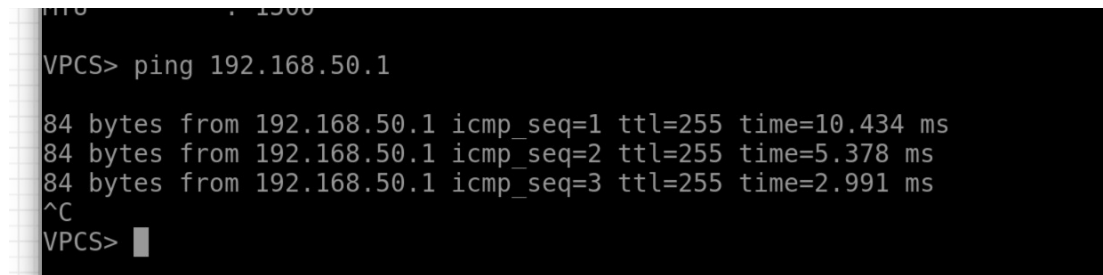
VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 192.168.50.10/24
GATEWAY    : 192.168.50.1
DNS        :
MAC        : 00:50:79:66:68:12
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS>
```

Gambar 12: IP PC LAN Huawei 192.168.50.10/24

```
1 ping 192.168.50.1
2
```



```
VPCS> ping 192.168.50.1
84 bytes from 192.168.50.1 icmp_seq=1 ttl=255 time=10.434 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=255 time=5.378 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=255 time=2.991 ms
^C
VPCS>
```

Gambar 13: Ping ke gateway 192.168.50.1 dari PC LAN Huawei berhasil

3. Konfigurasi IP address link Cisco ↔ Huawei Link 1 (network 10.10.10.0/30).

Cisco Router:

```
1 interface gi0/3
2   description LINK1-T0-HUAWEI
3   ip address 10.10.10.1 255.255.255.252
4   no shutdown
5 exit
6
```

Huawei Router:

```
1 interface Ethernet1/0/0
2   description LINK1-T0-CISCO
3   ip address 10.10.10.2 255.255.255.252
4   undo shutdown
5   quit
6 commit
7
```

4. Konfigurasi IP address link Cisco ↔ Huawei Link 2 (network 10.10.11.0/30).

Cisco Router:

```
1 interface gi0/0
2   description LINK2-T0-HUAWEI
3   ip address 10.10.11.1 255.255.255.252
4   no shutdown
5 exit
6
```

Huawei Router:

```
1 interface Ethernet1/0/4
2   description LINK2-T0-CISCO
3   ip address 10.10.11.2 255.255.255.252
4   undo shutdown
5   quit
6 commit
7
```

5. Konfigurasi IP address link Huawei ↔ MikroTik (network 10.10.20.0/30).

Huawei Router:

```
1 interface Ethernet1/0/3
2   description LINK-T0-MIKROTIK
3   ip address 10.10.20.1 255.255.255.252
4   undo shutdown
5   quit
6 commit
7
```

MikroTik:

```
1 /ip address add address=10.10.20.2/30 interface=ether1 comment=T0-HUAWEI
2
```

6. Konfigurasi IP address link Cisco ↔ MikroTik (network 10.10.30.0/30).

Cisco Router:

```
1 interface gi0/1
2   description LINK-T0-MIKROTIK
3   ip address 10.10.30.1 255.255.255.252
4   no shutdown
5 exit
6
```

MikroTik:

```
1 /ip address add address=10.10.30.2/30 interface=ether2 comment=T0-CISCO
2
```

7. Verifikasi konfigurasi IP address pada ketiga router.

Cisco Router (show ip interface brief):

```
Router#
Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  10.10.11.1      YES manual up          up
GigabitEthernet0/1  10.10.30.1      YES manual up          up
GigabitEthernet0/2  unassigned      YES unset up          up
GigabitEthernet0/2.10  192.168.10.1   YES manual up          up
GigabitEthernet0/2.20  192.168.20.1   YES manual up          up
GigabitEthernet0/2.30  192.168.30.1   YES manual up          up
GigabitEthernet0/2.40  192.168.40.1   YES manual up          up
GigabitEthernet0/3    10.10.10.1      YES manual up          up
Router#
Router#
```

Gambar 14: Verifikasi IP interface Cisco Router setelah konfigurasi antarrouter

Huawei Router (display ip interface brief):

```

Error: Incomplete command found at '^' position.
[~HUAWEI]
[~HUAWEI]quit
<HUAWEI>
<HUAWEI>display ip interface brief
*down: administratively down
!down: FIB overload down
^down: standby
(l): loopback
(s): spoofing
(d): Dampening Suppressed
(E): E-Trunk down
The number of interface that is UP in Physical is 23
The number of interface that is DOWN in Physical is 0
The number of interface that is UP in Protocol is 16
The number of interface that is DOWN in Protocol is 7

Interface                IP Address/Mask      Physical  Protocol VPN
Ethernet1/0/0             10.10.10.2/30         up        up        --
Ethernet1/0/0.4094        128.1.138.137/16     up        up        l3vpn
Ethernet1/0/1             unassigned            up        down      --
Ethernet1/0/1.4094        128.1.138.137/16     up        up        l3vpn
Ethernet1/0/2             192.168.50.1/24       up        up        --
Ethernet1/0/2.4094        128.1.138.137/16     up        up        l3vpn
Ethernet1/0/3             10.10.20.1/30         up        up        --
Ethernet1/0/3.4094        128.1.138.137/16     up        up        l3vpn
Ethernet1/0/4             10.10.11.2/30         up        up        --
Ethernet1/0/4.4094        128.1.138.137/16     up        up        l3vpn
Ethernet1/0/5             unassigned            up        down      --
----- More -----

```

Gambar 15: Verifikasi IP interface Huawei Router

MikroTik (/ip address print):

```

repeat new password>
[admin@MikroTik] > /ip address add address=10.10.20.2/30 interface=ether1 comment=TO-HUAWEI
[admin@MikroTik] > /ip address add address=10.10.30.2/30 interface=ether2 comment=TO-CISCO
[admin@MikroTik] > /ip address print
Flags: X - disabled, I - invalid, D - dynamic
# ADDRESS NETWORK INTERFACE
0 ;; TO-HUAWEI 10.10.20.2/30 ether1
1 ;; TO-CISCO 10.10.30.2/30 ether2
[admin@MikroTik] >

```

Gambar 16: Verifikasi IP address MikroTik

8. Verifikasi konektivitas antarrouter dengan melakukan ping ke router tetangga langsung.

Dari Cisco Router:

```

1 ping 10.10.10.2
2 ping 10.10.11.2
3 ping 10.10.30.2
4

```

```

GigabitEthernet0/5 10.10.10.1 PLS manual up up
Router#
Router#ping 10.10.10.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/5 ms
Router#ping 10.10.11.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 2/3/7 ms
Router#ping 10.10.30.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.30.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 1/2/3 ms
Router#

```

Gambar 17: Ping ke Huawei dan MikroTik dari Cisco Router berhasil

Dari Huawei Router:

```
1 ping 10.10.10.1
2 ping 10.10.11.1
3 ping 10.10.20.2
4
```

```
<HUAWEI>
<HUAWEI>ping 10.10.10.1
  PING 10.10.10.1: 56 data bytes, press CTRL_C to break
    Reply from 10.10.10.1: bytes=56 Sequence=1 ttl=255 time=2 ms
    Reply from 10.10.10.1: bytes=56 Sequence=2 ttl=255 time=3 ms
    Reply from 10.10.10.1: bytes=56 Sequence=3 ttl=255 time=3 ms
    Reply from 10.10.10.1: bytes=56 Sequence=4 ttl=255 time=3 ms
    Reply from 10.10.10.1: bytes=56 Sequence=5 ttl=255 time=2 ms

  --- 10.10.10.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 2/2/3 ms

<HUAWEI>
<HUAWEI>ping 10.10.11.1
  PING 10.10.11.1: 56 data bytes, press CTRL_C to break
    Reply from 10.10.11.1: bytes=56 Sequence=1 ttl=255 time=2 ms
    Reply from 10.10.11.1: bytes=56 Sequence=2 ttl=255 time=3 ms
    Reply from 10.10.11.1: bytes=56 Sequence=3 ttl=255 time=3 ms
    Reply from 10.10.11.1: bytes=56 Sequence=4 ttl=255 time=2 ms
    Reply from 10.10.11.1: bytes=56 Sequence=5 ttl=255 time=3 ms

  --- 10.10.11.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 2/2/3 ms

<HUAWEI>
<HUAWEI>ping 10.10.20.2
  PING 10.10.20.2: 56 data bytes, press CTRL_C to break
    Reply from 10.10.20.2: bytes=56 Sequence=1 ttl=64 time=2 ms
    Reply from 10.10.20.2: bytes=56 Sequence=2 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=3 ttl=64 time=1 ms
    Reply from 10.10.20.2: bytes=56 Sequence=4 ttl=64 time=2 ms
    Reply from 10.10.20.2: bytes=56 Sequence=5 ttl=64 time=1 ms

  --- 10.10.20.2 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 1/1/2 ms

<HUAWEI>
<HUAWEI>|
```

Gambar 18: Ping ke Cisco dan MikroTik dari Huawei Router berhasil

Dari MikroTik:

```
1 ping 10.10.20.1
2 ping 10.10.30.1
3
```

```
[admin@MikroTik] > ping 10.10.20.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.20.1                          56 255 4ms
1 10.10.20.1                          56 255 1ms
2 10.10.20.1                          56 255 1ms
3 10.10.20.1                          56 255 1ms
4 10.10.20.1                          56 255 1ms
5 10.10.20.1                          56 255 1ms
sent=6 received=6 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=4ms

[admin@MikroTik] > ping 10.10.30.1
SEQ HOST                                SIZE TTL TIME  STATUS
0 10.10.30.1                          56 255 2ms
1 10.10.30.1                          56 255 1ms
sent=2 received=2 packet-loss=0% min-rtt=1ms avg-rtt=1ms max-rtt=2ms

[admin@MikroTik] > |
```

Gambar 19: Ping ke Huawei dan Cisco dari MikroTik berhasil

1.4 Praktikum 4 — Konfigurasi OSPF Multivendor dan OSPF Cost

1. Konfigurasi OSPF pada Cisco Router dengan router-id 1.1.1.1, lalu iklankan semua network yang terhubung ke area 0.

```
1 router ospf 1
2 router-id 1.1.1.1
3 network 10.10.10.0 0.0.0.3 area 0
4 network 10.10.11.0 0.0.0.3 area 0
5 network 10.10.30.0 0.0.0.3 area 0
6 network 192.168.10.0 0.0.0.255 area 0
7 network 192.168.20.0 0.0.0.255 area 0
8 network 192.168.30.0 0.0.0.255 area 0
9 network 192.168.40.0 0.0.0.255 area 0
10 exit
11
```

Atur OSPF cost pada interface Cisco agar link ke Huawei menjadi jalur utama (cost 10) dan link ke MikroTik menjadi jalur backup (cost 100).

```
1 interface gi0/3
2 ip ospf cost 10
3 exit
4 interface gi0/0
5 ip ospf cost 10
6 exit
7 interface gi0/1
8 ip ospf cost 100
9 exit
10
```

Verifikasi network yang diiklankan OSPF.

```
1 show ip protocols
2
```



```

Gateway of last resort is not set

Router#
Router#show ip protocols
*** IP Routing is NSF aware ***

Routing Protocol is "application"
  Sending updates every 0 seconds
  Invalid after 0 seconds, hold down 0, flushed after 0
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Maximum path: 32
  Routing for Networks:
  Routing Information Sources:
    Gateway         Distance      Last Update
  Distance: (default is 4)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.10.10.0 0.0.0.3 area 0
    10.10.11.0 0.0.0.3 area 0
    10.10.30.0 0.0.0.3 area 0

Router#

```

Gambar 20: Verifikasi show ip protocols pada Cisco Router

2. Konfigurasi OSPF pada Huawei Router dengan router-id 2.2.2.2.

```

1 ospf 1 router-id 2.2.2.2
2 area 0.0.0.0
3 network 10.10.10.0 0.0.0.3
4 network 10.10.11.0 0.0.0.3
5 network 10.10.20.0 0.0.0.3
6 network 192.168.50.0 0.0.0.255
7 quit
8 quit
9

```

Atur OSPF cost pada interface Huawei.

```

1 interface Ethernet1/0/0
2 ospf cost 10
3 quit
4 interface Ethernet1/0/4
5 ospf cost 10
6 quit
7 interface Ethernet1/0/3
8 ospf cost 100
9 quit
10 commit
11

```

Verifikasi OSPF peer pada Huawei Router setelah konfigurasi awal dengan Cisco.

```

1 display ospf peer
2

```



```

[*HUAWEI]commit
[~HUAWEI]
[~HUAWEI]display ospf peer

(M) Indicates MADJ neighbor

      OSPF Process 1 with Router ID 2.2.2.2
      Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
  State: Full          Mode:Nbr is Slave      Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2      MTU: 1500
Dead timer due in 38 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
  State: Full          Mode:Nbr is Slave      Priority: 1
DR: 10.10.11.1        BDR: None             MTU: 1500
Dead timer due in 36 sec
Retrans timer interval: 5
Neighbor is up for 00h00m05s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

[~HUAWEI]
[~HUAWEI]
[~HUAWEI]

```

Gambar 21: OSPF peer Huawei dengan Cisco melalui dua link utama (state Full)

Verifikasi routing table OSPF Huawei.

```

1 display ip routing-table protocol ospf
2

```

```

[~HUAWEI]display ip routing-table protocol ospf
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
_public_ Routing Table : OSPF
Destinations : 9      Routes : 14

OSPF routing table status : <Active>
Destinations : 5      Routes : 10

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
-----
10.10.30.0/30       OSPF   10   110      D  10.10.11.1  Ethernet1/0/4
10.10.30.0/30       OSPF   10   110      D  10.10.10.1  Ethernet1/0/0
192.168.10.0/24      OSPF   10   11        D  10.10.11.1  Ethernet1/0/4
192.168.20.0/24      OSPF   10   11        D  10.10.10.1  Ethernet1/0/0
192.168.30.0/24      OSPF   10   11        D  10.10.11.1  Ethernet1/0/4
192.168.30.0/24      OSPF   10   11        D  10.10.10.1  Ethernet1/0/0
192.168.40.0/24      OSPF   10   11        D  10.10.11.1  Ethernet1/0/4
192.168.40.0/24      OSPF   10   11        D  10.10.10.1  Ethernet1/0/0

OSPF routing table status : <Inactive>
Destinations : 4      Routes : 4

Destination/Mask    Proto  Pre  Cost    Flags NextHop         Interface
-----
10.10.10.0/30       OSPF   10   10        10.10.10.2  Ethernet1/0/0
10.10.11.0/30       OSPF   10   10        10.10.11.2  Ethernet1/0/4
10.10.20.0/30       OSPF   10  100        10.10.20.1  Ethernet1/0/3
192.168.50.0/24     OSPF   10   1         192.168.50.1 Ethernet1/0/2

[~HUAWEI]
[~HUAWEI]

```

Gambar 22: Routing table OSPF Huawei Router

```

1 display ospf routing
2

```

```
[~HUAWEI]display ospf routing

      OSPF Process 1 with Router ID 2.2.2.2
        Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
10.10.10.0/30    10        Direct    10.10.10.2    2.2.2.2        0.0.0.0
10.10.11.0/30    10        Direct    10.10.11.2    2.2.2.2        0.0.0.0
10.10.20.0/30    100       Direct    10.10.20.1    2.2.2.2        0.0.0.0
10.10.30.0/30    110       Stub      10.10.11.1    1.1.1.1        0.0.0.0
10.10.30.0/30    110       Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.10.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.20.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.30.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.11.1    1.1.1.1        0.0.0.0
192.168.40.0/24  11        Stub      10.10.10.1    1.1.1.1        0.0.0.0
192.168.50.0/24  1         Direct    192.168.50.1  2.2.2.2        0.0.0.0

Total Nets: 9
Intra Area: 9 Inter Area: 0 ASE: 0 NSSA: 0

      OSPF Process 65534 with Router ID 128.1.138.137
        Routing Tables

Routing for Network
Destination      Cost      Type      NextHop      AdvRouter      Area
128.1.138.137/32 0         Direct    128.1.138.137 128.1.138.137 0.0.0.0

Total Nets: 1
Intra Area: 1 Inter Area: 0 ASE: 0 NSSA: 0
[~HUAWEI]
```

Gambar 23: OSPF routing table detail pada Huawei Router

3. Konfigurasi OSPF pada MikroTik dengan router-id 3.3.3.3 sebagai jalur backup.

```
1 /routing ospf instance set default router-id=3.3.3.3
2 /routing ospf network add network=10.10.20.0/30 area=backbone
3 /routing ospf network add network=10.10.30.0/30 area=backbone
4 /routing ospf interface add interface=ether1 cost=100
5 /routing ospf interface add interface=ether2 cost=100
6
```

```
[admin@MikroTik] > /routing ospf instance set default router-id=3.3.3.3
[admin@MikroTik] > /routing ospf network add network=10.10.20.0/30 area=backbone
[admin@MikroTik] > /routing ospf network add network=10.10.30.0/30 area=backbone
[admin@MikroTik] > /routing ospf interface add interface=ether1 cost=100
[admin@MikroTik] > /routing ospf interface add interface=ether2 cost=100
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=0.0.0.0 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0 adjacency=2s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=10s
[admin@MikroTik] > |
```

Gambar 24: Konfigurasi OSPF MikroTik dan verifikasi neighbor awal

4. Verifikasi OSPF neighbor pada ketiga router setelah semua konfigurasi selesai.

Cisco Router (show ip ospf neighbor):

```
Router#
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
3.3.3.3          1     FULL/BDR        00:00:37    10.10.30.2     GigabitEthernet0/1
2.2.2.2          1     FULL/BDR        00:00:39    10.10.11.2     GigabitEthernet0/0
2.2.2.2          1     FULL/BDR        00:00:33    10.10.10.2     GigabitEthernet0/3
Router#
Router#
```

Gambar 25: OSPF neighbor Cisco: Huawei muncul dua kali, MikroTik satu kali

Huawei Router (display ospf peer):

```
CISCO ROUTER  X  CISCO SWITCH  X  HUAWEI PCJ  X  HUAWEI ROUTER  X  MIKROTIK  X  +  v

OSPF Process 1 with Router ID 2.2.2.2
Neighbors

Area 0.0.0.0 interface 10.10.10.2 (Eth1/0/0)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.10.1
State: Full            Mode:Nbr is Slave  Priority: 1
DR: 10.10.10.1        BDR: 10.10.10.2    MTU: 1500
Dead timer due in 35 sec
Retrans timer interval: 5
Neighbor is up for 00h05m00s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.20.1 (Eth1/0/3)'s neighbors
Router ID: 3.3.3.3      Address: 10.10.20.2
State: Full            Mode:Nbr is Master Priority: 1
DR: 10.10.20.1        BDR: 10.10.20.2    MTU: 1500
Dead timer due in 32 sec
Retrans timer interval: 5
Neighbor is up for 00h00m53s
Neighbor Up Time : 2026-06-06 04:16:11
Authentication Sequence: [ 0 ]

Area 0.0.0.0 interface 10.10.11.2 (Eth1/0/4)'s neighbors
Router ID: 1.1.1.1      Address: 10.10.11.1
State: Full            Mode:Nbr is Slave  Priority: 1
DR: 10.10.11.1        BDR: 10.10.11.2    MTU: 1500
Dead timer due in 33 sec
Retrans timer interval: 5
Neighbor is up for 00h05m00s
Neighbor Up Time : 2026-06-06 04:12:04
Authentication Sequence: [ 0 ]

<HUAWEI>
```

Gambar 26: OSPF peer Huawei: Cisco via dua link, MikroTik satu link

MikroTik (/routing ospf neighbor print):

```
6 ADo 192.168.50.0/24 10.10.20.1 110
[admin@MikroTik] > /routing ospf neighbor print
0 instance=default router-id=2.2.2.2 address=10.10.20.1 interface=ether1 priority=1 dr-address=10.10.20.1
  backup-dr-address=10.10.20.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=1m20s
1 instance=default router-id=1.1.1.1 address=10.10.30.1 interface=ether2 priority=1 dr-address=10.10.30.1
  backup-dr-address=10.10.30.2 state="Full" state-changes=5 ls-retransmits=0 ls-requests=0 db-summaries=0
  adjacency=1m28s
[admin@MikroTik] >
```

Gambar 27: OSPF neighbor MikroTik: Huawei dan Cisco muncul

5. Verifikasi routing table OSPF pada Cisco Router.

```
1 show ip route 192.168.50.0
2
```

```
Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:50 ago
Routing Descriptor Blocks:
  10.10.11.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/0
    Route metric is 11, traffic share count is 1
  * 10.10.10.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/3
    Route metric is 11, traffic share count is 1
Router#
Router#
```

Gambar 28: Route menuju 192.168.50.0/24 via dua jalur utama Cisco-Huawei

```
1 show ip route ospf
2
```

```

CISCO ROUTER  x  CISCO SWITCH  x  HUAWEI PCJ  x  HUAWEI ROUTER  x  MIKROTIK  x  +
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:50 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
Router#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
O       10.10.20.0/30 [110/110] via 10.10.11.2, 00:06:05, GigabitEthernet0/0
        [110/110] via 10.10.10.2, 00:06:05, GigabitEthernet0/3
O       192.168.50.0/24 [110/11] via 10.10.11.2, 00:06:05, GigabitEthernet0/0
        [110/11] via 10.10.10.2, 00:06:05, GigabitEthernet0/3
Router#
Router#

```

Gambar 29: Semua route OSPF pada Cisco Router

Verifikasi routing table OSPF MikroTik.

```

1 /ip route print where ospf
2

```

```

[admin@MikroTik] > /ip route print where ospf
Flags: X - disabled, A - active, D - dynamic, C - connect, S - static, r - rip, b - bgp, o - ospf, m - mme,
B - blackhole, U - unreachable, P - prohibit
#   DST-ADDRESS      PREF-SRC  GATEWAY          DISTANCE
0   ADo  10.10.10.0/30          10.10.20.1         110
    10.10.30.1
1   ADo  10.10.11.0/30          10.10.20.1         110
    10.10.30.1
2   ADo  192.168.10.0/24         10.10.30.1         110
3   ADo  192.168.20.0/24         10.10.30.1         110
4   ADo  192.168.30.0/24         10.10.30.1         110
5   ADo  192.168.40.0/24         10.10.30.1         110
6   ADo  192.168.50.0/24         10.10.20.1         110
[admin@MikroTik] >

```

Gambar 30: Route OSPF pada MikroTik

6. Uji konektivitas *end-to-end* dari PC setiap VLAN menuju PC LAN Huawei (192.168.50.10).

Dari PC VLAN 10:

```
VLAN10
0;VPC4
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=254 time=7.133 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=254 time=5.788 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=254 time=4.602 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=254 time=5.584 ms
^C
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.519 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.501 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.377 ms
^C
VPCS> |
```

Gambar 31: Ping dari PC VLAN 10 ke PC LAN Huawei berhasil

Dari PC VLAN 20:

```
VLAN20
0;VPC5
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.065 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.257 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.831 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=8.408 ms
^C
VPCS> |
```

Gambar 32: Ping dari PC VLAN 20 ke PC LAN Huawei berhasil

Dari PC VLAN 30:

```
VLAN30
0;VPC6
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=6.550 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=7.185 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=7.030 ms
^C
VPCS> |
```

Gambar 33: Ping dari PC VLAN 30 ke PC LAN Huawei berhasil

Dari PC VLAN 40:

```

VLAN40
0;VPC7
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=7.291 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=6.187 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=6.197 ms
84 bytes from 192.168.50.10 icmp_seq=4 ttl=62 time=10.577 ms
^C
VPCS>

```

Gambar 34: Ping dari PC VLAN 40 ke PC LAN Huawei berhasil

7. Uji konektivitas dari PC LAN Huawei ke semua VLAN Cisco.

```

1 ping 192.168.10.11
2 ping 192.168.20.11
3 ping 192.168.30.10
4 ping 192.168.40.10
5

```

```

CISCO ROUTER  X  CISCO SWITCH  X  HUAWEI PC  X  HUAWEI ROUTER  X
OSPF  10  11  D  10.10.10.1  Ethernet1/
OSPF routing table status : <Inactive>
Destinations : 4  Routes : 4

Destination/Mask  Proto  Pre  Cost  Flags  NextHop  Interface
10.10.10.0/30  OSPF  10  10  10.10.10.2  Ethernet1/
10.10.11.0/30  OSPF  10  10  10.10.11.2  Ethernet1/
10.10.20.0/30  OSPF  10  100  10.10.20.1  Ethernet1/
192.168.50.0/24  OSPF  10  1  192.168.50.1  Ethernet1/
<HUAWEI>
<HUAWEI>
<HUAWEI>
<HUAWEI>ping 192.168.10.11
  PING 192.168.10.11: 56 data bytes, press CTRL_C to break
    Reply from 192.168.10.11: bytes=56 Sequence=1 ttl=63 time=4 ms
    Reply from 192.168.10.11: bytes=56 Sequence=2 ttl=63 time=4 ms
    Reply from 192.168.10.11: bytes=56 Sequence=3 ttl=63 time=4 ms
    Reply from 192.168.10.11: bytes=56 Sequence=4 ttl=63 time=5 ms
    Reply from 192.168.10.11: bytes=56 Sequence=5 ttl=63 time=4 ms

  --- 192.168.10.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 4/4/5 ms

<HUAWEI>
<HUAWEI>

```

Gambar 35: Ping dari PC LAN Huawei ke VLAN 10 berhasil


```

<HUAWEI>ping 192.168.20.11
  PING 192.168.20.11: 56 data bytes, press CTRL_C to break
    Reply from 192.168.20.11: bytes=56 Sequence=1 ttl=63 time=6 ms
    Reply from 192.168.20.11: bytes=56 Sequence=2 ttl=63 time=4 ms
    Reply from 192.168.20.11: bytes=56 Sequence=3 ttl=63 time=6 ms
    Reply from 192.168.20.11: bytes=56 Sequence=4 ttl=63 time=18 ms
    Reply from 192.168.20.11: bytes=56 Sequence=5 ttl=63 time=4 ms

  --- 192.168.20.11 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 4/7/18 ms

<HUAWEI>
<HUAWEI>

```

Gambar 36: Ping dari PC LAN Huawei ke VLAN 20 berhasil

```

<HUAWEI>ping 192.168.30.10
  PING 192.168.30.10: 56 data bytes, press CTRL_C to break
    Reply from 192.168.30.10: bytes=56 Sequence=1 ttl=63 time=6 ms
    Reply from 192.168.30.10: bytes=56 Sequence=2 ttl=63 time=4 ms
    Reply from 192.168.30.10: bytes=56 Sequence=3 ttl=63 time=4 ms
    Reply from 192.168.30.10: bytes=56 Sequence=4 ttl=63 time=4 ms
    Reply from 192.168.30.10: bytes=56 Sequence=5 ttl=63 time=6 ms

  --- 192.168.30.10 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 4/4/6 ms

<HUAWEI>
<HUAWEI>

```

Gambar 37: Ping dari PC LAN Huawei ke VLAN 30 berhasil

```

<HUAWEI>ping 192.168.40.10
  PING 192.168.40.10: 56 data bytes, press CTRL_C to break
    Reply from 192.168.40.10: bytes=56 Sequence=1 ttl=63 time=4 ms
    Reply from 192.168.40.10: bytes=56 Sequence=2 ttl=63 time=7 ms
    Reply from 192.168.40.10: bytes=56 Sequence=3 ttl=63 time=4 ms
    Reply from 192.168.40.10: bytes=56 Sequence=4 ttl=63 time=6 ms
    Reply from 192.168.40.10: bytes=56 Sequence=5 ttl=63 time=6 ms

  --- 192.168.40.10 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 4/5/7 ms

<HUAWEI>
<HUAWEI>

```

Gambar 38: Ping dari PC LAN Huawei ke VLAN 40 berhasil

1.5 Praktikum 5 — Pengujian Failover OSPF

1. **Kondisi normal: dua link Cisco–Huawei aktif.** Cek route menuju LAN Huawei dan lakukan traceroute untuk memastikan traffic melewati jalur langsung Cisco → Huawei.

```

1 show ip route 192.168.50.0
2 trace 192.168.50.10
3

```

```

Router#
Router#show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:05:50 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:05:50 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router#
Router#

```

Gambar 39: Route 192.168.50.0/24 saat dua link aktif: via dua next-hop Huawei

```

VLAN10
0;VPC4
VPCS> ping 192.168.50.1

84 bytes from 192.168.50.1 icmp_seq=1 ttl=254 time=7.133 ms
84 bytes from 192.168.50.1 icmp_seq=2 ttl=254 time=5.788 ms
84 bytes from 192.168.50.1 icmp_seq=3 ttl=254 time=4.602 ms
84 bytes from 192.168.50.1 icmp_seq=4 ttl=254 time=5.584 ms
^C
VPCS> ping 192.168.50.10

84 bytes from 192.168.50.10 icmp_seq=1 ttl=62 time=10.519 ms
84 bytes from 192.168.50.10 icmp_seq=2 ttl=62 time=8.501 ms
84 bytes from 192.168.50.10 icmp_seq=3 ttl=62 time=8.377 ms
^C
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    6.215 ms  3.868 ms  3.852 ms
 2  10.10.11.2     5.338 ms  5.235 ms  5.901 ms
 3  *192.168.50.10 8.304 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> |

```

Gambar 40: Traceroute kondisi normal: traffic via Cisco → Huawei langsung

- Kondisi 1 link Cisco–Huawei dimatikan.** Matikan interface Gi0/3 pada Cisco Router, lalu cek route dan traceroute kembali.

```

1 configure terminal
2 interface gi0/3
3 shutdown
4 exit
5

```

```

1 do show ip route 192.168.50.0
2 trace 192.168.50.10
3

```



```

Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:14:14 ago
  Routing Descriptor Blocks:
    * 10.10.11.2, from 2.2.2.2, 00:14:14 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
Router(config)#
Router(config)#

```

Gambar 41: Route 192.168.50.0/24 saat satu link mati: hanya tersisa satu next-hop via Gi0/0

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    5.552 ms  3.380 ms  4.727 ms
 2  10.10.11.2     7.359 ms  5.610 ms  4.944 ms
 3  *192.168.50.10 10.154 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> |

```

Gambar 42: Traceroute saat satu link mati: traffic tetap via jalur Cisco–Huawei yang tersisa

- Kondisi dua link Cisco–Huawei dimatikan (failover ke MikroTik).** Matikan juga interface Gi0/0, lalu cek route dan traceroute untuk memastikan traffic berpindah ke jalur backup melalui MikroTik.

```

1 configure terminal
2 interface gi0/0
3 shutdown
4 exit
5

```

```

1 do show ip route 192.168.50.0
2 trace 192.168.50.10
3

```

```

Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 201, type intra area
  Last update from 10.10.30.2 on GigabitEthernet0/1, 00:00:13 ago
  Routing Descriptor Blocks:
    * 10.10.30.2, from 2.2.2.2, 00:00:13 ago, via GigabitEthernet0/1
      Route metric is 201, traffic share count is 1
Router(config)#
Router(config)#

```

Gambar 43: Route 192.168.50.0/24 saat dua link mati: berpindah ke jalur backup via MikroTik (Gi0/1)

```

VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    3.614 ms  3.631 ms  4.510 ms
 2  10.10.30.2     4.652 ms  3.447 ms  3.378 ms
 3  10.10.20.1    11.464 ms  6.592 ms  5.997 ms
 4  *192.168.50.10  6.087 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> |

```

Gambar 44: Traceroute failover: jalur PC VLAN → Cisco → MikroTik → Huawei → LAN Huawei

4. **Mengaktifkan kembali dua link Cisco–Huawei.** Hidupkan kembali kedua interface dan verifikasi bahwa routing kembali ke jalur utama.

```
1 configure terminal
2 interface gi0/3
3   no shutdown
4 exit
5 interface gi0/0
6   no shutdown
7 exit
8
```

```
1 do show ip route 192.168.50.0
2 trace 192.168.50.10
3
```

```
Router(config)#do show ip route 192.168.50.0
Routing entry for 192.168.50.0/24
  Known via "ospf 1", distance 110, metric 11, type intra area
  Last update from 10.10.11.2 on GigabitEthernet0/0, 00:00:38 ago
  Routing Descriptor Blocks:
    10.10.11.2, from 2.2.2.2, 00:00:38 ago, via GigabitEthernet0/0
      Route metric is 11, traffic share count is 1
    * 10.10.10.2, from 2.2.2.2, 00:00:38 ago, via GigabitEthernet0/3
      Route metric is 11, traffic share count is 1
Router(config)#
Router(config)#
```

Gambar 45: Route 192.168.50.0/24 setelah link aktif kembali: dua next-hop via Huawei

```
VPCS> trace 192.168.50.10
trace to 192.168.50.10, 8 hops max, press Ctrl+C to stop
 1  192.168.10.1    4.942 ms  5.270 ms  4.586 ms
 2  10.10.11.2     7.938 ms  5.845 ms  5.319 ms
 3  *192.168.50.10 10.986 ms (ICMP type:3, code:3, Destination port unreachable)
VPCS> |
```

Gambar 46: Traceroute setelah pemulihan: traffic kembali via jalur utama Cisco–Huawei

2 Analisis Hasil Percobaan

Bagian ini membahas hasil yang diperoleh selama pelaksanaan praktikum modul 5, mencakup evaluasi terhadap lima percobaan yang dilakukan, yaitu konfigurasi Cisco Switch, konfigurasi Cisco Router, konfigurasi Huawei Router beserta IP address antarrouter, konfigurasi OSPF multivendor, dan pengujian failover OSPF.

2.1 Konfigurasi Cisco Switch

Konfigurasi Cisco Switch berjalan tanpa kendala. VLAN 10, 20, 30, dan 40 berhasil dibuat dan statusnya langsung aktif, yang terkonfirmasi dari output `show vlan brief`. Access port untuk masing-masing PC juga berhasil dikonfigurasi sesuai VLAN-nya, begitu pula trunk port ke Cisco Router pada interface Gi0/0 yang sudah membawa keempat VLAN tersebut. Dari output `show interfaces trunk`

terlihat bahwa encapsulation 802.1Q aktif dan VLAN 10, 20, 30, 40 sudah diizinkan melewati trunk, yang artinya switch sudah siap meneruskan traffic berlabel ke router.

2.2 Konfigurasi Cisco Router

Konfigurasi Cisco Router juga berjalan lancar. Metode ROAS (*Router on a Stick*) berhasil diterapkan dengan membuat subinterface Gi0/2.10 sampai Gi0/2.40 masing-masing sebagai gateway setiap VLAN. DHCP server untuk VLAN 10 dan 20 juga berfungsi dengan baik, terbukti dari PC VLAN 10 yang mendapat IP 192.168.10.11/24 dan PC VLAN 20 mendapat 192.168.20.11/24 secara otomatis. Sementara PC VLAN 30 dan 40 berhasil dikonfigurasi dengan IP statis sesuai tabel addressing.

Pengujian ping ke gateway dari masing-masing PC semuanya berhasil, yang menandakan subinterface dan enkapsulasi dot1Q sudah terkonfigurasi dengan benar. Uji ping antar-VLAN dari PC VLAN 10 ke VLAN 20, 30, dan 40 juga semua berhasil, membuktikan bahwa Cisco Router sudah berfungsi sebagai inter-VLAN router dengan benar.

2.3 Konfigurasi Huawei Router dan IP Address Antarrouter

LAN Huawei berhasil dikonfigurasi pada interface Ethernet1/0/2 dengan IP 192.168.50.1/24, dan PC LAN Huawei berhasil ping ke gateway tersebut. Konfigurasi IP pada empat link antarrouter juga semuanya berhasil, yakni dua link Cisco–Huawei, satu link Huawei–MikroTik, dan satu link Cisco–MikroTik. Dari hasil verifikasi `show ip interface brief`, `display ip interface brief`, dan `/ip address print`, seluruh interface sudah dalam kondisi up dengan IP yang sesuai.

Pengujian ping antarrouter juga berhasil di semua sisi. Cisco bisa ping ke Huawei dan MikroTik, Huawei bisa ping ke Cisco dan MikroTik, dan MikroTik bisa ping ke keduanya. Ini menjadi fondasi yang penting sebelum masuk ke konfigurasi OSPF, karena OSPF tidak akan bisa membentuk neighbor jika koneksi layer-3 di antara router belum berjalan.

2.4 Konfigurasi OSPF Multivendor dan OSPF Cost

Konfigurasi OSPF multivendor antara Cisco, Huawei, dan MikroTik berhasil dilakukan. Ketiga router berhasil membentuk adjacency di area 0. Dari output `show ip ospf neighbor` pada Cisco terlihat bahwa Huawei muncul dua kali (via Gi0/0 dan Gi0/3) dan MikroTik muncul satu kali, sesuai topologi. Huawei juga menampilkan tiga peer pada `display ospf peer`, dan MikroTik menampilkan dua neighbor.

Pengaturan OSPF cost berhasil membuat dua link Cisco–Huawei menjadi jalur utama (cost 10) dan link via MikroTik menjadi jalur backup (cost 100). Hal ini terkonfirmasi dari routing table Cisco yang menampilkan dua next-hop menuju 192.168.50.0/24 via 10.10.11.2 dan 10.10.10.2 dengan metric 11, sedangkan jalur via MikroTik tidak muncul karena cost-nya lebih tinggi.

Uji konektivitas end-to-end juga berhasil sepenuhnya. Semua PC dari VLAN 10, 20, 30, dan 40 bisa ping ke PC LAN Huawei di 192.168.50.10, dan sebaliknya PC LAN Huawei bisa ping ke semua VLAN Cisco. Ini membuktikan bahwa pertukaran informasi routing OSPF antarvendor berjalan dengan baik meski menggunakan tiga platform yang berbeda.

2.5 Pengujian Failover OSPF

Pengujian failover OSPF menghasilkan perilaku yang sesuai dengan yang diharapkan. Pada kondisi normal dengan dua link aktif, traceroute dari PC VLAN menuju LAN Huawei menunjukkan jalur langsung dua hop (gateway Cisco → Huawei), memanfaatkan salah satu dari dua link utama.

Ketika satu link Cisco–Huawei dimatikan (Gi0/3 di-shutdown), OSPF langsung menghapus satu next-hop dari routing table dan traffic tetap melewati link Cisco–Huawei yang tersisa via Gi0/0. Traceroute masih menunjukkan dua hop yang sama, hanya saja next-hop berubah ke 10.10.11.2. Perpindahan ini berlangsung cukup cepat tanpa intervensi manual sama sekali.

Saat kedua link Cisco–Huawei dimatikan, barulah OSPF memilih jalur backup via MikroTik. Routing table menunjukkan next-hop berpindah ke 10.10.30.2 (MikroTik) dengan metric 201, dan traceroute memperlihatkan jalur empat hop: gateway Cisco → MikroTik → Huawei → LAN Huawei. Ini membuktikan bahwa mekanisme failover OSPF berfungsi dengan baik sesuai desain cost yang sudah diatur sebelumnya.

Setelah kedua link diaktifkan kembali, routing table langsung kembali ke kondisi normal dengan dua next-hop via Huawei dan metric 11, dan traceroute kembali menunjukkan jalur dua hop langsung. Dari keseluruhan pengujian ini, bisa disimpulkan bahwa OSPF berhasil melakukan konvergensi secara otomatis baik saat link gagal maupun saat link pulih kembali.

3 Hasil Tugas Modul

Di push di github kelompok

4 Kesimpulan

Praktikum modul 5 ini mencakup konfigurasi jaringan multivendor yang melibatkan Cisco Switch, Cisco Router, Huawei Router, dan MikroTik dengan routing dinamis OSPF. Pada praktikum 1, Cisco Switch berhasil dikonfigurasi dengan empat VLAN (10, 20, 30, 40) beserta access port dan trunk port ke Cisco Router. Pada praktikum 2, Cisco Router dikonfigurasi menggunakan metode ROAS untuk menangani inter-VLAN routing, sekaligus sebagai DHCP server untuk VLAN 10 dan 20. Semua PC di tiap VLAN berhasil mendapatkan IP dan saling terhubung. Pada praktikum 3, seluruh link antarrouter berhasil dikonfigurasi dan diverifikasi, meliputi dua link Cisco–Huawei, satu link Huawei–MikroTik, dan satu link Cisco–MikroTik, serta LAN Huawei dengan network 192.168.50.0/24. Pada praktikum 4, OSPF berhasil dikonfigurasi pada ketiga vendor yang berbeda dalam satu area backbone. Pengaturan cost membuat dua link Cisco–Huawei menjadi jalur utama dan link via MikroTik menjadi jalur backup, dan seluruh PC lintas segmen bisa saling ping. Pada praktikum 5, failover OSPF terbukti berjalan otomatis. Saat satu link mati, traffic tetap lewat link Cisco–Huawei yang masih aktif. Saat kedua link mati, traffic berpindah ke jalur backup via MikroTik. Setelah link pulih, routing kembali normal tanpa konfigurasi ulang.

5 Lampiran

5.1 Dokumentasi saat praktikum

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3
10	FINANCE	active	
20	HR	active	
30	IT	active	
40	MARKETING	active	
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch#  
Switch#
```

Gambar 47: Gambar Saat Praktikum, Maaf Mas Gaada Foto Lain